facing rendezvous windows.

In terms of hardware to power the contraption, five silver/zinc-oxide batteries provided power after the CSM detached – three for re-entry and after landing and two for vehicle separation and parachute deployment. It also had twelve 420 N nitrogen tetroxide/hydrazine reaction control thrusters. Basically, it was the Command Module that provided the re-entry capability at the end of the mission after separation from the Service Module.

The Apollo 15 Command and Service Module as viewed from the Apollo Lunar Module

The Service Module itself was a cylinder measuring 3.9 metres in diameter and 7.6 metres long, which was attached to the back of the CM. The outer skin of the SM was formed of 2.5-centimetre-thick aluminium honeycomb panels. The interior was divided by milled aluminium radial beams into six sections around a central cylinder. At the back of the SM was a liquid propellant 91,000 N engine and cone shaped engine nozzle. Attitude control was provided by four identical banks of four 450 N reaction control thrusters, each spaced 90 degrees apart around the front.

The six sections of the SM held three 31-cell hydrogen oxygen fuel cells which provided 28 volts, two cryogenic oxygen and two cryogenic hydrogen tanks, four tanks for the main propulsion engine, two for fuel and two for oxidizer, and the subsystems the main propulsion unit. Two helium tanks were mounted in the central cylinder. Electrical power system radiators were at the top of the cylinder and environmental control radiator panels spaced around the bottom.

The Lunar Module – the last component of an Apollo spacecraft – was the part that would detach in lunar orbit and proceed to land on the moon. Since this aspect has to also do with lunar exploration, we'll take a look at it in the next chapter.